Is Chain of Thought an Interchangeable Scratchpad?

J-space ablation versus external reasoning across problem difficulty (Qwen3-4B)

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1 Hypothesis

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In the paper, the authors ablate the ten most active directions in the J-space at every token position across a band of middle layers, and find that GSM8K solved with a chain of thought (CoT) survives this ablation while the same problems answered directly do not. They conclude that the page and the J-space workspace are partially "interchangeable scratchpads." To interrogate this, we ask two questions. First, and as our main investigation, what is the relationship between the internal and external scratchpads? Second, how much protection does the CoT provide as problems harden?

For the first question, the authors suggest that CoT survives because the model "externalises" onto the page, yet they do not investigate what this externalisation means. We consider four mechanism hypotheses.

1. **(exchange)** externalisation actually *removes* the representation from the J-space, which is no longer needed to hold it.
2. **(duplication)** externalisation merely *duplicates* information that continues to be held in the J-space.
3. **(complementary)** the externalised and internal contents are complementary, a division of labour.
4. **(epiphenomenal)** the correlation is unrelated to storage altogether. Every generated token buys another forward pass, so the benefit could be extra serial computation rather than anything written down.

For the second question, we hypothesise **(H1)** that the protection offered by CoT is roughly constant across difficulty. This seems intuitive from the original paper as a genuinely interchangeable scratchpad that substitutes for the damaged workspace on grade-school problems should do the same on harder ones. We note in advance that the cross-dataset comparison can only weakly test this, since GSM8K, MATH-500 and AIME differ in topic, format and contamination as well as difficulty. We hypothesise our results will show the following, GSM8K at ∼90% with CoT and ∼25% direct with AIME direct at floor, a guess that the model can chain three to seven operations internally without a page (the serial budget D of §2) with behavioural collapse within ±2, and a floor rule, a dataset with zero clean direct accuracy cannot inform the comparison.

2 Experiment design

We first validate the pipeline by reproducing the original paper's Figure-24 contrast, crossing {CoT, direct} with {clean, light, medium, heavy ablation} on GSM8K. The reproduction succeeds (Fig. 1), with CoT essentially unaffected at any strength.

To test our 4 mechanistic hypotheses, we measure *occupancy*, how strongly the model is internally representing a token v at a given layer and position, the projection $\mathrm{occ}(v, \ell, t) = \langle h_{\ell,t}, \hat{\jmath}_v \rangle$ onto v's J-lens direction.

We investigate hypotheses (1, exchange) and (2, duplication) with a *maintenance test* of whether the model holds a value internally across a gap. The model reads a definition such as $x = 7$, then a stretch of unrelated text, then a question asking for the value. Occupancy alone is not decisive here, because the definition stays visible and the model can always re-read it rather than hold it, so we make re-reading impossible. An attention mask hides the definition from the question onward while the gap stays readable, so re-reading is blocked but carrying the value through the gap positions is not. A severed control masks the gap as well, so the value cannot get through at all, which sets the lower bound. A page condition keeps the mask but restates the value inside the gap, leaving a written copy readable, as a CoT's page would be. A model holding the value internally loses nothing when the definition is hidden. A model that answers by re-reading loses its only source and falls to the severed floor. We test hypothesis (1) a second way with a *dual-task probe*. The model holds an unrelated single-token word while solving, and we measure recall of the word afterwards and its occupancy during solving. If writing frees capacity, both should presumably be stronger with CoT than without. Next, a *correspondence test* allows us to investigate hypothesis (3, complementary). If the division of labour were by content kind, the workspace would hold content of a different kind from what a CoT writes, so it should not represent page-type values. We measure occupancy of the intermediate values of small arithmetic chains in two settings, answered directly, where nothing is externalised, and answered with the model's own chain of thought, the scenario the hypothesis is actually about, where we read each value's occupancy as the model writes it, during the later lines, and at the line that uses it. If the values appear in the workspace, it handles page-type content and the by-content reading fails. We investigate hypothesis (4, epiphenomenal) with a *prefix-rescue test*. We prefill an ablated model with its own correct reasoning, with content-free filler of identical token length, or with another problem's reasoning. If filler rescues as well as content, the benefit might be better compute in chain of thought, not externalisation of representations. Finally, to characterise the capacity CoT putatively replaces, a *depth ladder* of sequential single-digit chains (lengths 2–8, each line consuming the previous value) is answered without CoT, measuring accuracy and the serial budget D, the largest length whose answer still surfaces in the readout before emission.

For the second question, we repeated the validation experiment on MATH-500, stratified over its five levels, and both AIME exams.

For controls, we considered that our ablations could reflect any sufficiently large perturbation rather than the J-space specifically. A *norm-matched random control* removes an equal-norm component in random directions. A *rank-shifted control* removes lens ranks 51–60 instead of the true ablation's ranks 1–10. If damage came from removing lens directions as such rather than the most active ones, it would hurt as much. The paper's method also includes an exclusion filter, which never ablates a direction whose token the clean model already ranks in its top ten next-token predictions. In direct answering the very next token *is* the final answer, so this filter can shield the answer itself from ablation and distort the CoT-direct comparison, so we made it toggleable and checked the contrast without it.

3 Experimental details

We quantify damage with two ratios. *Retention* is ablated accuracy divided by clean accuracy within one answering mode, so a retention of 1.0 means the ablation cost nothing. The *protection ratio* $\rho = \mathrm{ret(CoT)}/\mathrm{ret(direct)}$ then compares the two modes, and is left undefined wherever clean direct accuracy sits at the floor.

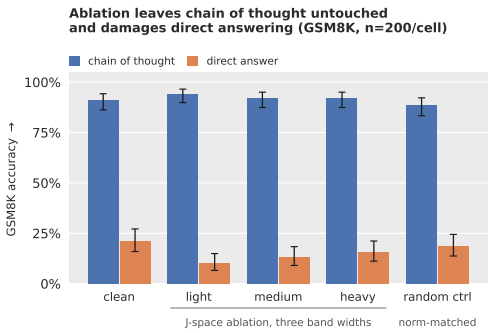


Figure 1: Reproducing the paper's Figure-24 contrast on Qwen3-4B (error bars are Wilson 95% CIs, n=200 per cell). Ablation at any band width leaves chain of thought untouched while damaging direct answering, and the norm-matched random control shows the damage requires the selected directions.

	clean		medium abl.		retention		
dataset	CoT	direct	CoT	direct	CoT	direct	ρ [95% CI]
GSM8K (n=200)	91.0%	21.0%	92.0%	13.0%	1.01	0.62	1.63 [1.05, 2.73]
MATH-500 (n=243)	65.0%	23.9%	67.9%	19.8%	1.04	0.83	1.26 [0.88, 1.84]
AIME (n=60)	23.3%	0.0%	18.3%	3.3%	0.79	–	undef. (floor)

Table 1: The difficulty axis. Retention is ablated accuracy over clean accuracy within one mode, so 1.0 means ablation cost nothing. AIME's ρ is undefined by the floor rule (clean direct 0/60). CoT cells on MATH-500 and AIME are budget-limited floors (124–135/243 and 59/60 truncated).

The reference implementation provides lens fitting and readout only, so the intervention itself is ours. At every position we score the vocabulary as $W_U(\text{norm}(J_\ell h))$, select the ten most active directions, and remove their span from the residual by least squares, with a paired clean forward pass supplying the exclusion filter. Our generation loop matches HuggingFace `generate()` token for token with ablation off, and our occupancy measure reproduces the reference readout's ordering (Spearman ≥ 0.994). We use the public pre-fitted Qwen3-4B Jacobian lens, and place the paper's bands by depth fraction at layers 22–26, 19–28 and 15–32 of 36. Unless stated otherwise, each cell contains n=200 problems (MATH-500 n=243, stratified, and AIME its full 60 problems) with one sample per problem at T=0.6 and seed 0. Answers are extracted from the generated text, with `math-verify` handling MATH and AIME equivalence. Accuracies carry Wilson 95% intervals and ratios carry bootstrap intervals. Generation is capped at 2048–4096 new tokens for CoT and 32 for direct answering, and exact dataset revisions are pinned in the repository. Our main statistical weakness is sampling depth, one sample and one seed outside the headline cells, which did replicate at a second seed (pooled $\rho = 1.50$ [1.13, 2.03]). One tokenizer fact shaped every probe. Qwen3 tokenizes numbers digit by digit, and multi-token values are exactly where the paper reports its readout degrading, so every value we ask the lens to read is a single token.

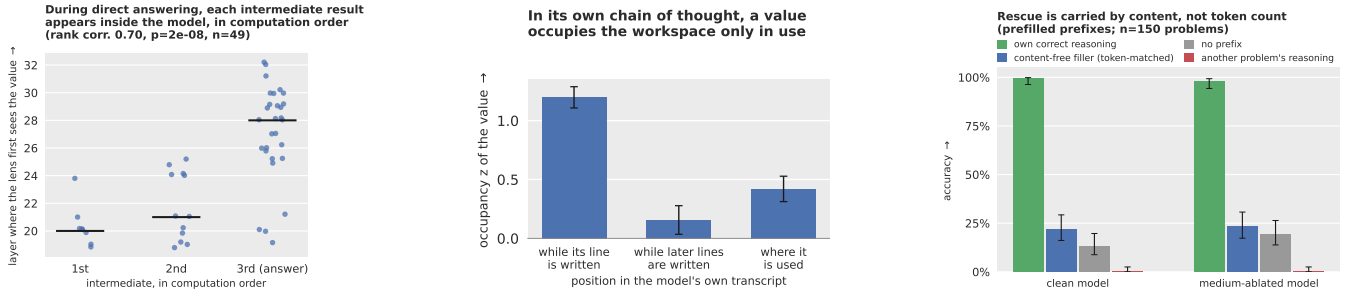
4 Experimental results

Replication with controls (Fig. 1, Table 1). The paper's contrast reproduces cleanly. Chain of thought is untouched at every band width (91.0% [86.2, 94.2] clean against 94.0/92.0/92.0% under light, medium and heavy ablation), while direct answering falls from 21.0% [15.9, 27.2] to 10.0–15.5%, giving $\rho = 1.63$ [1.05, 2.73]. The controls then localise the damage. A norm-matched random ablation and a rank-shifted ablation each retain 0.88 of clean performance where the true ablation retains 0.62, so the harm is specific to the currently most active J-lens directions rather than to perturbation as such. Removing the exclusion filter shifts ρ only from 1.63 to 1.57, so the mode contrast is not an artifact of the filter either.

Mechanism instruments (Fig. 2). In the maintenance test we found no internal holding. Once the definition is made unreadable at the point of use, accuracy collapses from 100% to 13.5% [8.1, 21.8], indistinguishable from the severed lower bound of 12.5% (n=96 per condition), while a restatement inside the gap restores 100%. The model does not carry even a single literal digit across a thirty-token gap. It answers by re-reading, and gap occupancy is near zero. At least for literal values at this scale, storage lives in the context, and with nothing held there is nothing for the page to duplicate and nothing for writing to remove, so hypotheses (1) and (2) both lose their premise. The dual-task probe gave a consistent result. Recall of the held word stayed at ceiling in all four conditions ($\geq$117/120), but recall only tests re-reading, since the word remains visible in the prompt, and the word's occupancy showed no asymmetry between modes, so writing freed no measurable capacity. In the correspondence test, when the model solves a three-step chain directly, the intermediate values surface in the workspace in computation order (Fig. 2a), so the workspace represents the values a chain of thought would write. The same holds when the model reasons on its own page. While it writes a value in its own chain of thought, that value is strongly present in the workspace (z=1.20, n=69), it fades over the following lines (0.16) and returns where the value is used again (0.42, Fig. 2b). The by-content reading therefore fails in both settings, and the storage picture from the maintenance test reappears, values are touched while written or used and not kept in between. Detectability of single digits is modest (AUC 0.67 [0.59, 0.74] against our 0.70 bar). Finally, in the prefix-rescue test (Fig. 2c), an ablated model handed back its own correct reasoning recovers to 98.0%, next to 99.3% for its clean counterpart, while filler of exactly the same length stays at the no-prefix baseline (23.3% against 19.3%) and another problem's reasoning collapses to zero. What the chain of thought contributes is therefore its content and not the forward passes its tokens buy, and the epiphenomenal reading fails.

The serial depth budget, or what the page replaces (Fig. 3a). With the depth ladder we investigate how many operations the model can chain internally before it needs the page. The answer is two. Accuracy on the sequential chains is perfect at two operations and collapses to 30% at three (n=30 per length), and the fraction of runs in which the answer surfaces in the readout before emission collapses at the same length (63% $\rightarrow$ 30%). Our advance guess of $D \in [3, 7]$ was therefore wrong, but the predicted coincidence between the readout collapse and the behavioural cliff held exactly.

Difficulty (Table 1). On MATH-500 the pattern repeats, with chain of thought losing nothing under ablation while direct answering is damaged, and ρ falls from 1.63 to 1.26 [0.88, 1.84]. On AIME the floor rule applies (clean direct 0/60), leaving CoT retention (0.79) as the only informative quantity. The falling ρ needs three qualifications before it can be read as weakening protection. The cross-dataset difference is not individually significant ($\Delta\rho = 0.40$ [-0.45, 1.42]). The fall at MATH-500 comes from the *direct* arm recovering (retention 0.62 $\rightarrow$ 0.83) rather than from CoT protection weakening (1.01 $\rightarrow$ 1.04), so any direct evidence of weakening rests on AIME alone. And the random controls on MATH-500 ($\rho_{\text{rand}} = 1.28$) and AIME are indistinguishable from the true ablation, so direction-specific protection is established only on GSM8K (random ρ 1.10 against



(a) (b) (c)

Figure 2: Mechanism results (error bars are Wilson 95% CIs over problems). **(a)** The correspondence test. The model answers three-step arithmetic chains directly, and each dot marks one intermediate value, placed by its position in the computation (x) and the layer where it first becomes readable (y). Later values appear at later layers, so the workspace computes, in order, exactly the values a chain of thought would have written. **(b)** The CoT arm of the correspondence test. Mean occupancy of the first value inside the model's own chain of thought, in the three windows named on the axis (bars are 95% CIs of the mean z). **(c)** The prefix-rescue test. Accuracy of the clean and medium-ablated models under each prefilled prefix. Only the model's own reasoning rescues the ablated model, filler of the same length does no better than no prefix, and each prefix acts the same without ablation.



(a) (b)

Figure 3: Capacity and scale (error bars are Wilson 95% CIs over problems). **(a)** The depth ladder. Programs of L chained single-digit operations (e.g. $a = 7 - 3$, then $b = a + 5$, then $c = b - 6$) are answered without chain of thought. Accuracy and pre-emission surfacing of the answer collapse together between L=2 and 3, so the internal budget is two operations. **(b)** The scale sweep. Retention under medium ablation at three model sizes. Direct answering grows less susceptible with scale while chain of thought is never hurt.

1.63). On the axis that actually measures protection, CoT retention, the data are consistent with H1.

Scale (Fig. 3b). We applied one fixed recipe, the band at the same depth fraction and each model read through its own lens, to three model sizes. Direct-mode retention rises steadily with size (0.43/0.62/1.04 at 1.7B/4B/8B) while chain of thought is never hurt. Taken literally the trend extrapolates to no damage at all at Sonnet scale, which contradicts the paper's own result, and the contradiction dissolves once we notice what is held fixed. The paper re-derived its band for each model where we keep one recipe, and a fixed ten directions is a smaller share of a wider residual stream. The panel therefore measures susceptibility to one recipe, not whether a workspace exists at each scale.

5 Analysis of results

We draw two conclusions, within the limits set out below. First, it is clear that a 4B model does possess a selectively ablatable workspace, answering the original paper's open question about whether small models had a J-space at all. However, our epiphenomenalism and complementarity hypotheses were rejected outright, and we find that the J-space does not contain a duplicate of externalised representations in the chain-of-thought approach. Therefore, it is most likely that the externalisation of representations means that the internal space no longer contains those representations as prominently as before, though this longitudinal change is yet to be investigated. Second, on difficulty, chain-of-thought protection did not measurably weaken across more difficult datasets. CoT retention stayed at 1.01 and 1.04 on GSM8K and MATH-500, with only the uncontrolled AIME cell dipping (0.79), consistent with H1, and the falling ρ decomposes into direct-arm changes rather than weakening protection.

Two limitations qualify these claims. Statistically, most cells carry one sample and one seed, and the random controls are indistinguishable from the true ablation on MATH-500 and AIME, so direction-specific protection is established only on GSM8K. Mechanistically, the storage probes used literal single-digit values read through a lens of modest sensitivity (AUC 0.67), and nothing in the maintenance gap gave the model a reason to carry the value, so that result shows no default holding rather than no capacity. All evidence comes from one model, so confident general claims would need larger n, more models, and adversarial probe variants. With more time we would pursue three directions. First, a properly powered version of the difficulty comparison, with many seeds per cell and a per-item model across mode, ablation and dataset, so any decline in protection is estimated rather than read off two points. Second, a maintenance test of what the model can hold rather than what it holds by default, warning the model that the definition will become unavailable and asking it to remember the value, so success requires internal carrying. Third, whether the balance between internal and external workspace shifts with the nature of the task, comparing problems that stress arithmetic, recall, planning or verbal reasoning, and harder instances of each, asking when the model leans hardest on the page.